Patient:		Order #:	
Account#:	LOC:	Collection D/T:	
MR#:	ATT DOC:	Received D/T:	
Age/Sex:	Req By:	Date of report:	
DOB:	Other Doc:	Sample type:	Brain

ADDENDUM

Final Diagnosis:

SEGA ventricles

FINAL INTEGRATED DIAGNOSIS:

A. Brain, lateral ventricle, foramen of Monroe, tumor resection:

Integrated diagnosis: Low grade neuroepithelial tumor with *ST13- ROS1* fusion and copy number variations (partial 6q, whole 9, whole X, whole 22) plus *MUTYH* mutation

Histologic diagnosis: Low grade neuroepithelial tumor in keeping with Subependymal giant cells astrocytoma

Histologic Grade: WHO Grade I

Molecular information: *ST13- ROS1* fusion and copy number variations (partial 6q, whole 9, whole X, whole 22) plus *MUTYH* mutation

COMMENT:

The molecular evaluation of this tumor DOES NOT SUPPORT the association with tuberous sclerosis syndrome.
The tumor does not demonstrate *TSC1* or *TSC2* sequence variants. However rare cases of SEGA have been reported outside tuberous sclerosis. The molecular features are NOT specific for any particular morphologically identifiable tumor. The differential diagnosis includes chordoid glioma of the third ventricle that does not have any specific genetic mutation. However the age, the location and the negative stain for TTF-1 renders this possibility unlikely.

PRELIMINARY DIAGNOSIS

B. Brain, lateral ventricle, foramen of Monroe, tumor resection:
Integrated diagnosis: Pending
Histologic diagnosis: Low grade neuroepithelial tumor in keeping with Subependymal giant cells astrocytoma
Histologic grade: WHO Grade I

Molecular information: Pending

C. Brain, lateral ventricle, foramen of Monroe, tumor resection:
Blood clot, small, organizing

COMMENT
This is a low grade neuroepithelial tumor that primarily developed within the constellation of the tuberous complex. However occasional sporadic cases without other abnormalities of syndrome were described and reported. The differential diagnosis includes chordoid glioma of the third ventricle; however the location, the age and the negative stain for TTF-1 renders this possibility unlikely. The molecular evaluation of the tumor is pending and will reported in the FINAL INTEGRATED DIAGNOSIS.
Further tests and correlation with clinical history is recommended to exclude the possibility of tuberous sclerosis.

SYNOPTIC REPORT
History of Previous Tumor/Familial Syndrome:
None

Specimen Type/Procedure:
Total gross resection

Specimen Handling:
Frozen section
Permanent paraffin sections
Tissue for CBTTC
Frozen for genetic studies

Specimen Size:
See gross description

Laterality:
Left

Tumor Site:
Ventricle, Foramen Monroe

Histologic Type and Grade:
Subependymal giant cell astrocytoma

Histologic Grade:
WHO Grade I

Ancillary Studies:
Genetic tests

MICROSCOPIC EXAMINATION

Microscopic examination was performed on H&E stained sections and additional sections stained with immunoperoxidase methods for vimentin, GFAP, neurofilament, synaptophysin, S-100, CD34, EMA, INI-1, TTF-1 and Mib-1/Ki-67.
Microscopic sections demonstrate a tumor with an overall fascicular architecture, composed of spindle cells. In areas the tumor cells are plump with abundant glassy, pink cytoplasm. Some cells are multiple, eccentrically positioned nuclei. There are no mitoses, necrosis or microvascular proliferation. All tumor cells are diffusely and strongly reactive for vimentin and S-100. INI-1 is retained and TTF-1 negative. The positivity for GFAP is diffuse and only rare cells are positive for neurofilament. Synaptophysin, EMA and CD34 are more difficult to interpret but there are positive cells on a high background. CD34 highlights all vessels. The proliferative index is very low
The histologic and immunohistochemical features are most consistent with subependymal giant cells tumor.

FROZEN SECTION DIAGNOSIS
Specimen Description
Tumor present. Deferred to permanent
Reported by:
Reported to:
Date Reported:
Time Reported:

ADDENDUM

Next generation sequencing (NGS) analysis of genes in the Comprehensive Solid Tumor NGS Panel performed on the DNA and RNA extracted from the tumor specimen of the patient showed the following results:

1. One Tier 2 fusion gene: *ST13- ROS1*

2. Multiple Tier 2 copy number variations (CNVs)
NGS analysis performed on the DNA extracted from the tumor specimen identified multiple copy number variations (CNVs).
Significant findings are loss of partial 6q, including a contiguous portion of the ROS1 gene 5' to exon 35, which is consistent with the presence of the fusion gene described above.
Additionally, mosaic allelic imbalance (loss vs. copy neutral loss of heterozygosity) of whole chromosomes 9 (including TSC1), 22, and X is present.
Please note that the exact nature (loss versus loss of heterozygosity) of some of the above alterations cannot be accurately determined due to technical limitations of the assay.

3. One Tier 2 sequence variant: *MUTYH* (NM_012222.2), c.725G>A (p.Arg242His)

The presence of multiple copy number variants and a novel *ST13-ROS1* fusion gene is consistent with a neoplastic process. To the best of our knowledge, the *ST13-ROS1* fusion has not been reported and the clinical significance of the *ST13-ROS1* fusion is yet to be determined. However, *ROS1* gene fusions have been described in other tumors with response to targeted therapy (see full report on

Epic). Mosaic allelic imbalance of whole chromosome 9 is seen (including *TSC1*), **however NO *TSC1* or *TSC2* sequence variants were identified in this tumor within the detection limits of this assay.**

The complete report, signed out by Dr. , can be found on Epic ( ).

More immunostains were performed and the tumor is negative for cytokeratin, desmin, myogenin and MyoD1.

Medical History:

Preoperative Diagnosis: BRAIN TUMOR
Postoperative Diagnosis: PENDING
Procedure: CRANI
Clinical History: 6 Y/O FEMALE WITH BRAIN TUMOR; HERE FOR ABOVE PROCEDURE

Tissue:

Specimen: A Brain
brain tumor

Specimen: B Brain
brain tumor

Gross:

A: The specimen is received fresh for frozen section in a container labeled with the patient's name, medical record number and designated "lateral and 3rd ventricular tumor". It consists of a 2.0 x 2.0 x 1.4 cm aggregate of multiple tan-pink focally hemorrhagic soft tissue fragments. Portion of tissue is submitted for frozen section as FS A1 and subsequently submitted to DGD study ( ). One squash preps and one H&E stained slide are prepared. Additional portion of tissue is submitted to CBTTC (3x snap frozen, 1x DMSO).

The remaining tissue is submitted entirely in one cassette labeled A1. Also received are two tubes of blood and submitted to CBTTC.

B: The specimen is received fresh in a container labeled with the patient's name, medical record number and designated "lateral and 3rd ventricular tumor". It consists of a pink-red irregular soft tissue fragment measuring 0.4 x 0.3 x 0.1 cm. The specimen is submitted entirely in one cassette labeled B1.

Electronically Signed

"These tests were developed and their performance characteristics determined by the Pathology Department at . They have not been cleared or approved by the U.S. Food and Drug Administration. The FDA has determined that such clearance or approval is not necessary. These tests are used for clinical purposes. It should not be regarded as investigational or for research. This Laboratory is certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA) as qualified to perform high complexity clinical laboratory testing."